

$$\begin{aligned}
 & d) A'B'C'D + A'CD + B'CD + A'BCD + BC'D \\
 & = A'B'C'D + A'CD \cdot 1 + B'CD \cdot 1 + A'BCD + BC'D \cdot 1 \quad (\text{Postulate 2(b)}) \\
 & = A'B'C'D + A'CD \cdot (B+B') + B'CD \cdot (A+A') + A'BCD + BC'D \cdot (A+A') \quad (\text{Postulate 3(a)}) \\
 & = A'B'C'D + (A'CD \cdot B + A'CD \cdot B') + (B'CD \cdot A + B'CD \cdot A') + A'BCD + (BC'D \cdot A + BC'D \cdot A') \quad (\text{Postulate 4(a)})
 \end{aligned}$$

Rearranging the terms and removing repeated terms:

$$A'B'C'D + A'B'CD + A'BCD + A'BCD + AB'C'D + AB'CD + ABC'D + ABC'D$$

$$\Sigma(0, 2, 5, 7, 8, 10, 12, 13)$$

Simplified: $B'D + A'BD + ABC'$

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11	1	1		
10	1			1

$$\begin{aligned}
 & e) x'z + w'xy + w(x'y + xy') \\
 & = x'z + w'xy + (w(x'y + xy')) \quad (\text{Postulate 4(a)}) \\
 & = x'z + w'xy + (wxy + wxy') \quad (\text{Theorem 4(b)}) \\
 & = x'z \cdot 1 + w'xy + (wxy + wxy') \quad (\text{Postulate 2(b)}) \\
 & = x'z \cdot (w + w') + w'xy + (wxy + wxy') \quad (\text{Postulate 5(a)}) \\
 & = (x'zw + x'zw') + w'xy + (wxy + wxy') \quad (\text{Postulate 4(a)}) \\
 & = (x'zw \cdot 1 + x'zw' \cdot 1) + w'xy \cdot 1 + (wxy \cdot 1 + wxy' \cdot 1) \quad (\text{Postulate 2(b)}) \\
 & = (x'zw \cdot (y + y') + x'zw' \cdot (y + y')) + w'xy \cdot (z + z') + (wxy \cdot (z + z') + wxy' \cdot (z + z')) \quad (\text{Postulate 5(a)}) \\
 & = ((x'zwy + x'zwy') + (x'zwy + x'zwy')) + (w'xyz + w'xyz') + (wxyz + wxyz') \quad (\text{Postulate 4(a)})
 \end{aligned}$$

Rearranging the terms and removing repeated terms:

$$w'x'yz + w'x'yz + w'xyz + w'xyz + w'xyz + w'xyz + w'xyz + w'xyz$$

$$\Sigma(1, 3, 4, 5, 9, 10, 11, 12, 13)$$